

# EULER-VOIGT EQUATIONS

## 1. INTRODUCTION TESTS

We study the Euler-Voigt equations posed on a periodic domain  $\mathbb{T}^3 := (\mathbb{R}/\mathbb{Z})^3$

$$(1.1) \quad \begin{aligned} -\alpha^2 \partial_t \Delta \mathbf{u}_\alpha + \partial_t \mathbf{u}_\alpha + (\mathbf{u} \cdot \nabla) \mathbf{u}_\alpha + \nabla p &= 0, \\ \nabla \cdot \mathbf{u}_\alpha &= 0, \\ \mathbf{u}_\alpha(\mathbf{x}, 0) &= \boldsymbol{\eta}_\alpha, \end{aligned}$$

where  $\alpha > 0$  is a regularization parameter. The Euler-Voigt equations can be viewed as an inviscid regularization of the Euler equations

$$(1.2) \quad \begin{aligned} \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p &= 0, \\ \nabla \cdot \mathbf{u} &= 0, \\ \mathbf{u}(\mathbf{x}, 0) &= \boldsymbol{\eta}. \end{aligned}$$

The solutions to (1.2) and (1.1) both possess the following time scaling symmetry

$$(1.3) \quad \tilde{\mathbf{u}}(\mathbf{x}, t) = \lambda \mathbf{u}(\mathbf{x}, \lambda t).$$

Unlike the Euler equations for which the global well-posedness of the solutions remains an open problem, the solutions of the Euler-Voigt are known to be globally well-posed for  $t \in (-\infty, \infty)$  [1]. Moreover, the “ $\alpha$ -energy” of the solutions of (1.1) is conserved for all times

$$(1.4) \quad E_\alpha(t) := \|\mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = E_\alpha(0).$$

The behavior of the solutions of (1.1) as  $\alpha \rightarrow 0$  can be used to determine whether the solutions of the Euler equations will develop a finite time singularity [2].

**Theorem 1.1.** *Let  $\boldsymbol{\eta}_\alpha \in H^s$ ,  $s \geq 3$  with  $\nabla \cdot \boldsymbol{\eta}_\alpha = 0$  and let  $\mathbf{u}_\alpha$  be the corresponding unique solution of (1.1). Suppose that there exists  $T^* > 0$  such that*

$$(1.5) \quad \limsup_{\alpha \rightarrow 0^+} \left( \alpha \sup_{t \in [0, T^*]} \|\nabla \mathbf{u}_\alpha(t)\|_{L^2} \right) > 0.$$

*Then the Euler equations with the initial condition  $\boldsymbol{\eta}_\alpha$  must develop a singularity within the interval  $[0, T^*]$ .*

The Theorem 5.1 in [3] stated that

**Theorem 1.2.** *Given initial data  $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$  for some  $s \geq 3$ , let  $\mathbf{u}, \mathbf{u}_\alpha \in C([0, T], H^s(\mathbb{T}^3)) \cap C^1([0, T], H^{s-1}(\mathbb{T}^3))$  be the corresponding solutions to (1.2) and (1.1), respectively, where  $0 < T < T_{\max}$  and  $T_{\max}$  is the maximal time for which a solution to (1.2) exists and is unique. For all  $t \in [0, T]$ , we have*

$$(1.6) \quad \begin{aligned} &\|\mathbf{u}(t) - \mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla(\mathbf{u}(t) - \mathbf{u}_\alpha(t))\|_{L^2}^2 \\ &\leq \left( \|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \right) e^{Ct} + C\alpha^2(e^{Ct} - 1). \end{aligned}$$

Based on Theorem 1.2, we know that if

$$(1.7) \quad \|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \rightarrow 0, \quad \text{as } \alpha \rightarrow 0,$$

then for all  $t \in [0, T]$ ,

$$(1.8) \quad \|\mathbf{u}(t) - \mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla(\mathbf{u}(t) - \mathbf{u}_\alpha(t))\|_{L^2}^2 \rightarrow 0.$$

Therefore, we can put the constraint on  $\|\nabla \boldsymbol{\eta}_\alpha\|_{L^2}$  in the optimization process and check whether  $\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 \rightarrow 0$  as  $\alpha \rightarrow 0$  in computation.

We slightly modify Theorem 5.2 in [3] as follows.

**Theorem 1.3.** *Given initial data  $\eta, \eta_\alpha \in H^s(\mathbb{T}^3)$  for some  $s \geq 3$  satisfying*

$$(1.9) \quad \|\eta - \eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla(\eta - \eta_\alpha)\|_{L^2}^2 \rightarrow 0, \quad \text{as } \alpha \rightarrow 0,$$

*let  $u, u_\alpha \in C([0, T], H^s(\mathbb{T}^3)) \cap C^1([0, T], H^{s-1}(\mathbb{T}^3))$  be the corresponding solutions to (1.2) and (1.1), respectively.*

*If there exists a finite time  $T^* > 0$  such that the solutions  $u_\alpha$  satisfies*

$$(1.10) \quad \limsup_{\alpha \rightarrow 0} \left( \alpha^2 \sup_{t \in [0, T^*]} \|\nabla u_\alpha(t)\|_{L^2}^2 \right) > 0,$$

*then  $u$  develops a singularity in the interval  $[0, T^*]$ .*

*Proof.* Suppose that the solution  $u$  of the Euler equations (1.2) with the initial condition  $\eta$  is regular on  $[0, T^*]$ . Since the Euler-Voigt equations are globally well-posed, we have

$$(1.11) \quad \|u_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla u_\alpha(t)\|_{L^2}^2 = \|\eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \eta_\alpha\|_{L^2}^2,$$

thus

$$(1.12) \quad \alpha^2 \|\nabla u_\alpha(t)\|_{L^2}^2 = \|\eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \eta_\alpha\|_{L^2}^2 - \|u_\alpha(t)\|_{L^2}^2.$$

Taking the supremum over  $t \in [0, T^*]$  of the above equation, we obtain that

$$(1.13) \quad \sup_{t \in [0, T^*]} \alpha^2 \|\nabla u_\alpha(t)\|_{L^2}^2 = \|\eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \eta_\alpha\|_{L^2}^2 - \inf_{t \in [0, T^*]} \|u_\alpha(t)\|_{L^2}^2.$$

Taking the lim sup when  $\alpha \rightarrow 0$  of the above equation, we obtain that

$$(1.14) \quad \begin{aligned} \limsup_{\alpha \rightarrow 0} \sup_{t \in [0, T^*]} \alpha^2 \|\nabla u_\alpha(t)\|_{L^2}^2 &= \limsup_{\alpha \rightarrow 0} \left( \|\eta_\alpha\|_{L^2}^2 + \alpha^2 \|\nabla \eta_\alpha\|_{L^2}^2 - \inf_{t \in [0, T^*]} \|u_\alpha(t)\|_{L^2}^2 \right) \\ &= \|\eta\|_{L^2}^2 - \|u(t)\|_{L^2}^2, \end{aligned}$$

where the second equality holds because according to Theorem 1.2, the convergence of the solutions of the Euler-Voigt equations to the solution of the Euler equations is uniform on  $[0, T^*]$ . Since the energy of regular solutions of the Euler equations is conserved, we have  $\|\eta\|_{L^2}^2 = \|u(t)\|_{L^2}^2$ , thus

$$(1.15) \quad \limsup_{\alpha \rightarrow 0} \left( \sup_{t \in [0, T^*]} \alpha^2 \|\nabla u_\alpha(t)\|_{L^2}^2 \right) = 0,$$

which contradicts the hypothesis in the theorem that the above expression is positive.  $\square$

## 2. PDE OPTIMIZATION METHOD

**2.1. The optimization problem.** We consider analytic initial conditions that belong to the following Gevrey class

$$(2.1) \quad G^\sigma := \left\{ v \in H^s(\mathbb{T}^3) : \|v\|_{G^\sigma} = \langle v, v \rangle_{G^\sigma}^{1/2} < \infty \right\}, \quad s \geq 3, \quad \sigma > 0,$$

with the inner product

$$(2.2) \quad \begin{aligned} \forall v, u \in G^\sigma, \quad \langle v, u \rangle_{G^\sigma} &:= \sum_{j \in \mathbb{Z}^3} (1 + |2\pi j|^2)^s e^{4\pi\sigma|j|} \hat{v}_j \cdot \bar{\hat{u}}_j \\ &= \int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma|D|} v \cdot u \, dx, \end{aligned}$$

where overbar denotes complex conjugation, whereas the operators  $|D|$  and  $e^{\sigma|D|}$  are defined via

$$(2.3) \quad \left[ \widehat{|D|v} \right]_j := 2\pi|j|\hat{v}_j, \quad \left[ \widehat{e^{\sigma|D|}v} \right]_j := e^{2\pi\sigma|j|}\hat{v}_j.$$

Since the initial conditions in (1.1) and (1.2) need to be divergence-free and the quantity  $\int_{\mathbb{T}} \boldsymbol{\eta} \, dx$  is an invariant of motion, we introduce the following subspace in which optimal initial conditions will be sought

$$(2.4) \quad V := \{v \in G^\sigma : \nabla \cdot v = 0, \quad \int_{\mathbb{T}^3} v \, dx = \mathbf{0}\}.$$

According to the analytical results in Section 1, we define the objective functional  $\Phi_{\alpha,T} : V \rightarrow \mathbb{R}^+$  as

$$(2.5) \quad \Phi_{\alpha,T}(\boldsymbol{\eta}) := \|\nabla \mathbf{u}_\alpha(T; \boldsymbol{\eta})\|_{L^2}^2 = \|\mathbf{u}_\alpha(T; \boldsymbol{\eta})\|_{\dot{H}^1}^2,$$

where  $\mathbf{u}_\alpha(T; \boldsymbol{\eta})$  is the solution of the Euler-Voigt equations (1.1) obtained with the initial condition  $\boldsymbol{\eta} \in V$  and  $\alpha$  is the regularization parameter in (1.1).

Due to the scaling property (1.3) of the Euler-Voigt equations (1.1), we can restrict our discussion to initial conditions with unit  $\dot{H}^1$  seminorm which belong to a closed manifold  $\mathcal{M}_1 \subset V$  defined as

$$(2.6) \quad \mathcal{M}_1 = \{v \in V : \|v\|_{\dot{H}^1} = 1\}.$$

Therefore, we arrive at the following optimization problem

**Problem 2.1.** Given  $\alpha, T \in \mathbb{R}_+$ , find

$$(2.7) \quad \tilde{\boldsymbol{\eta}}_{\alpha,T} = \operatorname{argmax}_{\boldsymbol{\eta} \in \mathcal{M}_1} \Phi_{\alpha,T}(\boldsymbol{\eta}).$$

Our goal is to first fix  $T$  and solve the optimization problem 2.1 for a sequence of decreasing values of  $\alpha$  that converge to 0.

**2.2. The adjoint method.** We compute the gradient  $\nabla \Phi_{\alpha,T}(\boldsymbol{\eta})$  of the objective functional in (2.5) using an adjoint method which depends on solving a properly defined adjoint system backward in time. We first introduce the Gâteaux (directional) differential  $\Phi'_{\alpha,T}(\boldsymbol{\eta}; \boldsymbol{\eta}') : V \times V \rightarrow \mathbb{R}$

$$(2.8) \quad \begin{aligned} \Phi'_{\alpha,T}(\boldsymbol{\eta}; \boldsymbol{\eta}') &:= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [\Phi_{\alpha,T}(\boldsymbol{\eta} + \epsilon \boldsymbol{\eta}') - \Phi_{\alpha,T}(\boldsymbol{\eta})] \\ &= 2 \langle \mathbf{u}_\alpha(\cdot, T), \mathbf{u}'_\alpha(\cdot, T) \rangle_{\dot{H}^1}. \end{aligned}$$

Here  $\mathbf{u}'_\alpha(\mathbf{x}, t)$  is the solution of the linearization of the Euler-Voigt equations (1.1) around its solution  $\mathbf{u}_\alpha(\mathbf{x}, t; \boldsymbol{\eta})$ , which is defined by

$$(2.9) \quad \begin{aligned} \mathcal{L} \begin{bmatrix} \mathbf{u}'_\alpha \\ p'_\alpha \end{bmatrix} &:= \begin{bmatrix} \partial_t \mathbf{u}'_\alpha + (I - \alpha^2 \Delta)^{-1} (\mathbf{u}'_\alpha \cdot \nabla \mathbf{u}_\alpha + \mathbf{u}_\alpha \cdot \nabla \mathbf{u}'_\alpha + \nabla p'_\alpha) \\ \nabla \cdot \mathbf{u}'_\alpha \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix}, \\ \mathbf{u}'_\alpha(\mathbf{x}, 0) &= \boldsymbol{\eta}'(\mathbf{x}), \end{aligned}$$

where  $\boldsymbol{\eta}'$  is the perturbation of the initial condition and  $p'_\alpha(\mathbf{x}, t)$  is the corresponding pressure perturbation. We further introduce the adjoint states  $\{\mathbf{u}_\alpha^* : \mathbb{T}^3 \times [0, T] \rightarrow \mathbb{R}^3, p_\alpha^* : \mathbb{T}^3 \times [0, T] \rightarrow \mathbb{R}\}$  and the following duality-pairing relation

$$(2.10) \quad \begin{aligned} \left( \mathcal{L} \begin{bmatrix} \mathbf{u}'_\alpha \\ p'_\alpha \end{bmatrix}, \begin{bmatrix} \mathbf{u}_\alpha^* \\ p_\alpha^* \end{bmatrix} \right) &:= \int_0^T \int_{\mathbb{T}^3} \mathcal{L} \begin{bmatrix} \mathbf{u}'_\alpha \\ p'_\alpha \end{bmatrix} \cdot \begin{bmatrix} \mathbf{u}_\alpha^* \\ p_\alpha^* \end{bmatrix} \, dx \, dt \\ &= \left( \begin{bmatrix} \mathbf{u}'_\alpha \\ p'_\alpha \end{bmatrix}, \mathcal{L}^* \begin{bmatrix} \mathbf{u}_\alpha^* \\ p_\alpha^* \end{bmatrix} \right) + \int_{\mathbb{T}^3} \mathbf{u}'_\alpha(\mathbf{x}, T) \cdot \mathbf{u}_\alpha^*(\mathbf{x}, T) \, dx \\ &\quad - \int_{\mathbb{T}^3} \boldsymbol{\eta}'(\mathbf{x}) \cdot \mathbf{u}_\alpha^*(\mathbf{x}, 0) \, dx \\ &= 0. \end{aligned}$$

Here “.” denotes the usual Euclidean inner product on  $\mathcal{R}^4$  and  $\mathcal{L}^*$  is the *adjoint* operator defined in terms of the following system, which is obtained by performing integration by parts with respect to both space and time in (2.10)

$$(2.11) \quad \mathcal{L}^* \begin{bmatrix} \mathbf{u}_\alpha^* \\ p_\alpha^* \end{bmatrix} := \begin{bmatrix} -\partial_t \mathbf{u}_\alpha^* - (\nabla(I - \alpha^2 \Delta)^{-1} \mathbf{u}_\alpha^* + (\nabla(I - \alpha^2 \Delta)^{-1} \mathbf{u}_\alpha^*)^T) \mathbf{u}_\alpha - \nabla p_\alpha^* \\ -\nabla \cdot \mathbf{u}_\alpha^* \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix},$$

$$\mathbf{u}_\alpha^*(\mathbf{x}, T) = 2|D|^2 \mathbf{u}_\alpha(\mathbf{x}, T).$$

Therefore, we have

$$(2.12) \quad \Phi'_{\alpha, T}(\boldsymbol{\eta}; \boldsymbol{\eta}') = 2 \langle \mathbf{u}_\alpha(\cdot, T), \mathbf{u}'_\alpha(\cdot, T) \rangle_{\dot{H}^1} = \langle \mathbf{u}_\alpha^*(\cdot, 0), \boldsymbol{\eta}'(\cdot) \rangle_{L^2}$$

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